

PRODUCT REQUIREMENTS DOCUMENT

WhatsApp Leads Tracking System (WLTS)

An end-to-end platform to capture, auto-respond to, nurture, track and bill WhatsApp leads generated from Meta and Google Ads campaigns.

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Version History

Version	Date	Author	Summary of Changes
0.1	10-Sep-2026	Rahul Parmar	Initial notes and concept capture
1.0	12-Sep-2026	Rahul Parmar	First draft PRD for stakeholder review

Reviewers & Approvers

Name	Role	Approval Status
Rahul Parmar	Founder / Product Owner	Pending
Engineering Lead	Technical Feasibility	Pending
Design Lead	UX Review	Pending

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1. Executive Summary

WhatsApp Leads Tracking System (WLTS) is a SaaS platform that helps businesses capture leads generated from Instagram, Meta and Google ad campaigns that redirect into WhatsApp, and manage the entire lifecycle of that lead — from the first automated reply through nurturing, qualification, human hand-off, conversion and billing.

Businesses running ad campaigns today receive a high volume of WhatsApp conversations daily. Most teams cannot respond fast enough to every conversation, and there is no structured way to separate high-intent leads from low-quality ones, track the history of a conversation, or see what stage a lead is in. WLTS solves this by combining WhatsApp Business Platform (official API) connectivity, an automated reply and nurturing engine, a purpose-built CRM, and integrated billing into one system.

Each business ("Account") is onboarded by a WLTS admin, connected to its own WhatsApp number via the official WhatsApp Business Platform, and configured with reply flows, nurturing sequences and hand-off rules suited to that business.

2. Problem Statement

Businesses advertising on Instagram and Google frequently route interested prospects to a WhatsApp number as the point of first contact. This creates the following problems:

- High volume: Daily ad spend generates a continuous stream of WhatsApp conversations that outpaces the team's ability to reply manually.
- Mixed lead quality: Genuinely interested leads sit in the same inbox as spam, curiosity clicks and unqualified enquiries, with no automated way to separate them.
- Slow response time: Delayed replies to hot leads directly reduce conversion — prospects move on or lose interest.
- No structured tracking: WhatsApp alone has no CRM concept — there is no lead status, ownership, follow-up schedule or conversion tracking.
- No unified billing: Once a lead converts to a customer, invoicing and billing happen outside the conversation, in a disconnected tool or manual process.
- No visibility for the owner: Business owners cannot easily see communication history, where a conversation stalled, or overall team performance across all leads.

3. Goals & Objectives

The primary goal of WLTS is to maximize lead conversion from WhatsApp by combining automation with a clear, trackable handover to human sales effort — through an interface that is simple enough for non-technical business owners to configure and use.

3.1 Business Objectives

- Increase lead-to-customer conversion rate for client businesses.
- Reduce average first-response time on inbound WhatsApp leads to near-zero via automation.
- Give business owners full visibility into every lead's communication history and current status.
- Provide a single system that covers lead capture, nurturing, CRM tracking and billing — removing the need for disconnected tools.

3.2 Product Objectives (V1)

- Connect a business's WhatsApp number via the official WhatsApp Business Platform (BSP) API.
- Automatically reply to and nurture new leads using configurable message flows / chatbot logic.
- Let each business define the trigger point at which a lead is handed off from automation to a human (e.g., when a lead shows buying intent).
- Track every lead in a CRM with status, ownership, history and AI-generated remarks on conversation progress.
- Generate invoices and manage billing for converted customers, with Zoho Billing as the V1 billing engine.
- Provide analytics and reporting on lead volume, response time, conversion and revenue.

4. Target Users & Personas

Persona	Description	Primary Need
WLTS Admin (Anthropic-side)	Internal team operating the WLTS platform itself.	Create and onboard business accounts, monitor platform health, manage subscriptions.
Business Owner / Account Admin	Owner or manager of a business running ad campaigns into WhatsApp (e.g., D2C brand, agency, service provider).	See all leads and conversations, configure automation rules, track conversions and revenue.
Sales / Support Agent	Team member handling hand-off conversations for a business account.	Receive qualified leads with full context, reply quickly, update lead status.
End Customer / Lead	Person who clicked an Instagram/Google ad and messaged the business on WhatsApp.	Get a fast, relevant response and a smooth path to purchase — not a direct system user.

5. Scope

5.1 In Scope — Version 1 (MVP)

- Account & user onboarding by WLTS admin, including WhatsApp number connection.
- WhatsApp Business Platform (official API / BSP) integration for sending and receiving messages.
- Automated reply and lead-nurturing message flows (rule-based chatbot).
- Configurable hand-off rules to route a lead from automation to a human agent.
- CRM: lead capture, lead profile, status pipeline, ownership, notes.
- Full conversation history per lead, visible to authorized users (owner/admin), with AI-generated remarks summarizing where a conversation stands or stalled.
- Billing & invoicing for converted leads, integrated with Zoho Billing.
- Analytics dashboard: lead volume, response times, conversion rate, revenue.
- Role-based access control (Admin, Business Owner, Agent).

5.2 Out of Scope — Version 1

- Channels other than WhatsApp (Instagram DM, Facebook Messenger, SMS, email) — noted as future phase.
- Native ad-campaign creation/management (Meta/Google Ads) — WLTS consumes leads that arrive via WhatsApp, it does not manage ad spend.
- AI voice calling or voice-note transcription.
- Multi-currency / multi-language billing beyond what Zoho Billing supports out of the box.
- Native mobile apps (V1 is web-based, responsive).

6. User Roles & Permissions

Role	Key Permissions
WLTS Admin	Create/suspend business accounts; run onboarding; connect WhatsApp numbers to accounts; view platform-wide usage; no access to a business's lead conversations unless granted for support.
Business Owner (Account Admin)	Full access within their own account: configure automation & hand-off rules, manage agents, view all leads and full history, view analytics, manage billing/invoices.
Sales / Support Agent	Access to assigned leads and conversations, update lead status and notes, cannot change account-level configuration or billing.
End Customer (Lead)	Not a system user; interacts only via WhatsApp. No login.

7. Product Flow Overview

The end-to-end flow, from ad click to billed customer, is as follows:

1. A prospect clicks an Instagram or Google ad configured to open a WhatsApp conversation with the business's connected number.
2. The inbound message is captured by WLTS via the WhatsApp Business Platform API and a new Lead record is created in the CRM automatically.
3. The automated reply / nurturing flow engages the lead immediately (business hours or after-hours message, qualifying questions, information sharing).
4. Based on the business's configured hand-off rules, WLTS evaluates lead intent. If the lead meets the criteria (e.g., asks pricing, shares requirement, requests a callback), the conversation is flagged and forwarded to a human agent/owner.
5. If the lead does not show sufficient intent, it remains in the nurturing sequence, with scheduled automated follow-ups.
6. All communication — automated and human — is logged against the Lead's history, along with an AI-generated remark describing conversation status (e.g., "awaiting customer reply", "price objection raised", "chat stalled after quotation").
7. The agent/owner continues the conversation manually within WLTS (or directly on WhatsApp, synced back) until the lead converts or is marked lost.
8. On conversion, the owner generates an invoice from within WLTS; billing is processed and tracked through the Zoho Billing integration.
9. Analytics aggregate lead volume, response time, conversion rate and revenue for the account.

8. Functional Requirements

8.1 Onboarding & Account Management

ID	Requirement	Priority
FR-1.1	WLTS Admin can create a new business account with company details and a designated Business Owner login.	Must
FR-1.2	WLTS Admin can connect a business's WhatsApp number via the official WhatsApp Business Platform onboarding flow (Embedded Signup or equivalent).	Must
FR-1.3	WLTS Admin can configure account-level settings on behalf of the business during onboarding (working hours, initial reply flow, hand-off rule defaults) or hand this control to the Business Owner.	Must
FR-1.4	Business Owner can invite and manage Agent users within their account.	Must

ID	Requirement	Priority
FR-1.5	WLTS Admin can suspend, reactivate or delete a business account.	Should

8.2 WhatsApp Integration (Official API)

ID	Requirement	Priority
FR-2.1	System integrates with an official WhatsApp Business Platform tech provider (BSP) to send and receive messages per connected account.	Must
FR-2.2	System supports WhatsApp message templates for outbound messages sent outside the 24-hour customer service window, per Meta policy.	Must
FR-2.3	System receives inbound messages, media (images, documents, audio) and delivery/read status via webhooks in near real time.	Must
FR-2.4	System handles multiple connected WhatsApp numbers, one per business account, in isolation from one another.	Must

8.3 Automated Reply & Lead Nurturing (Chatbot)

ID	Requirement	Priority
FR-3.1	Business Owner can configure an automated first-reply message triggered on a new inbound lead.	Must
FR-3.2	Business Owner can build a nurturing sequence: a series of automated follow-up messages sent on a defined schedule if the lead has not converted or been handed off.	Must
FR-3.3	Chatbot supports basic conversational logic: qualifying questions, button/list replies, keyword-based branching.	Must
FR-3.4	Nurturing sequence pauses automatically once a lead is handed off to a human agent.	Must
FR-3.5	Business Owner can pause, edit or stop a nurturing sequence for an individual lead manually.	Should

8.4 Hand-off / Lead Forwarding Rules

ID	Requirement	Priority
FR-4.1	Business Owner can define one or more hand-off triggers (e.g., specific keywords, specific chatbot answers, a lead reaching a defined step) at which the lead is escalated from automation to manual handling.	Must

ID	Requirement	Priority
FR-4.2	On hand-off, the system notifies the assigned agent/owner (in-app and/or email/push) with full lead context.	Must
FR-4.3	Business Owner can configure lead-routing rules to assign forwarded leads to a specific agent or a round-robin pool.	Should
FR-4.4	A lead can be manually escalated to human hand-off at any time by an agent, overriding automation.	Must

8.5 CRM — Lead Management & Tracking

ID	Requirement	Priority
FR-5.1	Every inbound WhatsApp conversation automatically creates or updates a Lead record (name, number, source, first contact time).	Must
FR-5.2	Leads move through a configurable status pipeline (e.g., New → Engaged → Qualified → Handed Off → Converted / Lost).	Must
FR-5.3	Each lead record shows source attribution where available (e.g., campaign/ad reference passed via WhatsApp deep link).	Should
FR-5.4	Agents/owners can add manual notes, tags and next-follow-up-date reminders to a lead.	Must
FR-5.5	Leads can be filtered and searched by status, owner, source, date range and tag.	Must

8.6 Communication History & AI Remarks

ID	Requirement	Priority
FR-6.1	Every message (automated and human, inbound and outbound) is stored against the lead in full chronological history.	Must
FR-6.2	Authorized users (Business Owner, assigned Agent) can view the complete chat history for a lead within WLTS.	Must
FR-6.3	System generates an AI-written remark/summary per lead indicating current conversation status and where it left off (e.g., "Customer asked for pricing on 10-Sep, no reply since").	Should
FR-6.4	Business Owner has an account-wide communication overview showing recent activity across all customers/leads.	Should

8.7 Billing & Invoicing

ID	Requirement	Priority
FR-7.1	Owner can convert a Lead to a Customer and generate an invoice directly from the lead record.	Must
FR-7.2	Billing and invoicing operations are processed through an integration with Zoho Billing.	Must
FR-7.3	Invoice status (paid/unpaid/overdue) is visible on the customer's record within WLTS.	Must
FR-7.4	System supports recurring billing for repeat/subscription-based customers where applicable.	Could

8.8 Analytics & Reporting

ID	Requirement	Priority
FR-8.1	Dashboard shows lead volume over time, by source and by status.	Must
FR-8.2	Dashboard shows average first-response time and average time-to-conversion.	Must
FR-8.3	Dashboard shows conversion rate and revenue, filterable by date range and agent.	Must
FR-8.4	Reports can be exported (CSV/PDF).	Should

9. Non-Functional Requirements

Category	Requirement
Usability	Interface must be usable by non-technical business owners with minimal training; simple, uncluttered UI/UX.
Performance	Inbound WhatsApp messages must be processed and, where automated, replied to within a few seconds.
Scalability	System must support multiple business accounts and a growing volume of concurrent conversations without degradation.
Reliability	Webhook processing must be resilient to WhatsApp API retries and avoid duplicate lead creation or duplicate replies.
Security	Multi-tenant data isolation between business accounts; encrypted storage of customer data and access tokens.
Compliance	Adherence to WhatsApp Business Platform policy (24-hour session window, template approval) and applicable data protection regulations.

Category	Requirement
Auditability	All configuration changes (automation rules, hand-off rules) are logged with user and timestamp.

10. System Architecture Overview (High Level)

WLTS is composed of the following high-level components:

- WhatsApp Business Platform (official BSP/tech provider) — handles message transport in and out of WhatsApp.
- Ingestion & Webhook Layer — receives inbound messages/status events and normalizes them into internal events.
- Automation / Chatbot Engine — executes configured reply and nurturing flows, evaluates hand-off rules.
- CRM Core — stores leads, customers, statuses, notes, ownership and pipeline logic.
- Conversation Store — persists full message history per lead/customer, feeds the AI remarks generator.
- AI Remarks Service — periodically summarizes conversation state per lead using conversation content.
- Billing Integration Layer — synchronizes customer and invoice data with Zoho Billing.
- Analytics & Reporting Layer — aggregates events for dashboards and exports.
- Admin Console — WLTS-admin-only tool for account creation, onboarding and WhatsApp number connection.
- Business Web App — the primary interface for Business Owners and Agents (leads, chat, config, billing, analytics).

11. Key Data Entities (Overview)

Entity	Key Attributes
Account	Account ID, business name, connected WhatsApp number, plan/subscription status, working hours, created date.
User	User ID, account ID, name, role (Admin/Owner/Agent), contact details, status.
Lead / Customer	Lead ID, account ID, phone number, name, source/campaign, status, owner (agent), tags, converted flag.
Conversation / Message	Message ID, lead ID, direction (in/out), sender type (bot/agent/customer), content, timestamp, delivery status.

Entity	Key Attributes
Automation Rule	Rule ID, account ID, trigger type, condition, action (reply/hand-off/tag), sequence order.
Invoice	Invoice ID, lead/customer ID, Zoho reference ID, amount, status (paid/unpaid/overdue), due date.
AI Remark	Remark ID, lead ID, generated summary text, generated timestamp.

12. Third-Party Integrations

Integration	Purpose
WhatsApp Business Platform (official tech provider / BSP)	Send/receive WhatsApp messages, manage message templates, connect client WhatsApp numbers.
Zoho Billing	Generate, send and track customer invoices and recurring billing.
Meta / Google Ads (indirect)	Source of the WhatsApp click-to-chat leads; no direct API dependency in V1 beyond capturing referral/campaign parameters where WhatsApp exposes them.

13. Success Metrics (KPIs)

Metric	Target (Illustrative — to be finalized)
Average first-response time	Under 1 minute for all inbound leads (via automation)
Lead-to-conversion rate	Measurable uplift vs. business's pre-WLTS baseline
% of leads correctly auto-qualified vs. requiring manual correction	≥ 85%
Agent adoption / active daily usage	≥ 80% of onboarded agents active weekly
Invoice generation turnaround	Invoice raised within same session as conversion

14. Assumptions & Constraints

Assumptions

- Businesses already run or plan to run Instagram/Google ad campaigns that click through to WhatsApp.
- Each business account maps to one primary WhatsApp Business number in V1.

- Businesses are willing to have their WhatsApp number managed through an official BSP rather than the standard WhatsApp Business App.

Constraints

- Outbound messaging outside the 24-hour customer service window must use pre-approved WhatsApp message templates, per Meta policy.
- Billing functionality in V1 is dependent on Zoho Billing's API capabilities and account setup per business.
- WhatsApp Business Platform onboarding requires business verification with Meta, which is outside WLTS's direct control and may add lead time.

15. Risks & Mitigations

Risk	Mitigation
WhatsApp policy changes (templates, session window, pricing) affect automation behavior.	Abstract messaging logic behind an internal service layer; monitor Meta policy updates closely.
Over-automation frustrates leads or delays genuine hand-off.	Configurable, transparent hand-off rules; allow manual override by agents at any time.
AI remarks misclassify conversation status, misleading agents.	Treat AI remarks as an assistive summary, not a source of truth; always show full raw history alongside.
Multi-tenant data leakage between business accounts.	Strict account-scoped data access checks at the API layer; regular access audits.
Zoho Billing integration downtime blocks invoicing.	Queue invoice requests and retry; allow manual invoice status entry as fallback.

16. Release Phases

Phase	Focus
Phase 1 — MVP	WhatsApp connection, automated reply & nurturing, hand-off rules, core CRM, chat history, basic analytics, Zoho Billing integration.
Phase 2	AI remarks/summarization at scale, advanced routing (round-robin/skills-based), richer analytics, export/reporting.
Phase 3	Additional channels (Instagram DM, Messenger), deeper ad-platform attribution, mobile apps.

17. Open Questions

- Should one business account support multiple WhatsApp numbers (e.g., for different regions/brands) in V1 or later?
- What is the exact set of default hand-off triggers we ship out-of-the-box vs. fully custom per business?
- Is Zoho Billing the long-term billing engine, or an initial integration pending a native billing module?
- How is campaign/source attribution captured reliably when a lead arrives via WhatsApp click-to-chat (ref parameter availability)?
- What data retention policy applies to lead conversation history, and does it vary by region/regulation?

18. Glossary

Term	Definition
BSP	Business Solution Provider — an official WhatsApp technology provider used to access the WhatsApp Business Platform API.
Lead	A prospect who has initiated or been routed into a WhatsApp conversation with the business.
Hand-off	The point at which a conversation moves from automated handling to a human agent.
Nurturing	Scheduled automated follow-up messaging intended to keep a lead engaged until conversion or hand-off.
AI Remark	A short, system-generated summary describing the current state of a lead's conversation.